



SUPPLEMENTARY SEMESTER EXAMINATIONS - OCTOBER 2023

Program : **B.E. - Artificial Intelligence and Machine Learning / Artificial Intelligence and Data Science**
Course Name : **Theory of Computation**
Course Code : **AI44 / AD44**

Semester : **IV**
Max. Marks : **100**
Duration : **3 Hrs**

Instructions to the Candidates:

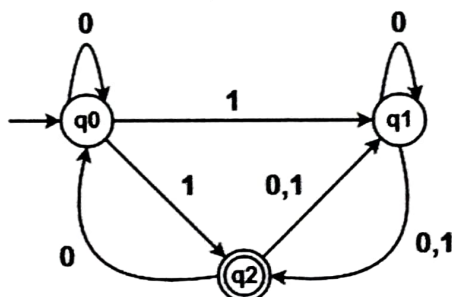
- Answer one full question from each unit.

UNIT - I

- Define the following terminologies with an example for each:
(i) Alphabet (ii) Power of an alphabet (iii) Symbol (iv) String (v) Concatenation. CO1 (10)
 - Convert the following ϵ -NFA to its equivalent DFA. CO1 (10)

	ϵ	a	b	c
$\rightarrow P$	{q,r}	$\emptyset$	{q}	{r}
q	$\emptyset$	{p}	{r}	{p,q}
*r	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

- Convert the following Non-Deterministic Finite State Machine (NFSM) to Deterministic Finite State Machine (DFSM). CO1 (10)



- Define DFSM. Design a DFSM to accept each of the following Language: CO1 (10)
 - $L = \{w \in \{a,b\}^*; w \text{ has all strings that end with substring } abb\}$
 - $L = \{w \in \{0,1\}^*; w \text{ has even number of } a's \text{ and even number of } b's\}$

UNIT - II

- Write Regular Expressions for the following language:- CO2 (04)
 - $L = \{uvw \mid u,w \text{ belong to } \{a,b\}^* \text{ and } |v|=2\}$
 - Set of strings over $\{a,b,c\}$ in which at least one a appears in the first three symbols.
 - Consider the following DFA:- CO2 (09)

States	0	1
$\Rightarrow q1$	q2	q1
q2	q3	q1
*q3	q3	q2

Find $R^{(3)}_{13}$. Using Kleen's theorem.

- Prove that if L is a regular language, so is L^R . CO4 (07)

4. a) Convert DFA to Regular expressions by eliminating states.

CO3 (07)

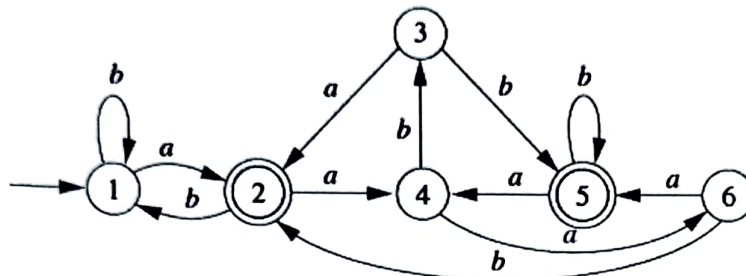
State	a	b
$\rightarrow^* q_0$	q1	q3
q1	q0	q2
q2	q3	q1
q3	q2	q0

- b) States and Prove Pumping Lemma for Regular Languages.

CO4 (05)

- c) Minimize the following DFA by applying table filling algorithm:-

CO3 (08)



UNIT - III

5. a) Define Ambiguity. Consider the grammar $E \rightarrow E+E|E^*E|(E)|id$
 (i) Find the leftmost, rightmost derivations and parse trees for the string $id+id*id$

CO3 (10)

(ii) Show that this grammar is ambiguous.

- b) Define Context Free Grammar. Simplify the grammar by removing ϵ - productions and unreachable symbols.

CO3 (10)

$S \rightarrow AB|AC$

$A \rightarrow aAb| \epsilon$

$B \rightarrow bA$

$C \rightarrow bCa$

$D \rightarrow AB$

6. a) Define Pushdown Automata(PDA) and design PDA to accept the language.

CO3 (08)

$$L(m) = \{\omega C \omega^R | \omega \in (A \cup B)^* \text{ and } \omega^R \text{ is the inverse of } \omega\}$$

- b) Write Context Free Grammar to specify:

CO3 (12)

(i) All string over (a, b) that are even and odd length and palindromes.

(ii) $L = \{a^n b^{2n} \text{ over } \sum = \{a, b\}, n \geq 1\}$

(iii) $L = \{0^m 1^m 2^n; m \geq 1, n \geq 0\}$

UNIT- IV

7. a) Prove the theorem: If L is a CFL and R is a regular language, then $L \cap R$ is a CFL.

CO4 (08)

- b) Prove that $L = \{0^n 1^n 2^n | n \geq 1\}$ is not a CFL.

CO4 (06)

- c) Consider the grammar:

CO3 (06)

$S \rightarrow ABC$

$A \rightarrow BC | a$

$B \rightarrow bAC | \epsilon$

$C \rightarrow cAB | \epsilon$

Identify all the nullable variables and eliminate ϵ - productions.

8. a) Begin with the grammar: CO3 (12)
 $S \rightarrow ABC \mid BaB$
 $A \rightarrow aA \mid BaC \mid aaa$
 $B \rightarrow bBb \mid a \mid D$
 $C \rightarrow CA \mid AC$
 $D \rightarrow \epsilon$
 i) Eliminate ϵ - productions.
 ii) Eliminate any unit productions in the resulting grammar.
 iii) Eliminate any useless symbols in the resulting grammar.
 iv) Put the resulting grammar into Chomsky Normal Form.
 b) State and prove pumping lemma for CFLs. CO4 (08)

UNIT - V

9. a) Define Turing Machine. CO5 (14)
 For $L = \{a^n b^n c^n, n \geq 1\}$ design a Turing Machine. Give the computational sequence for the input aabbcc.
 b) Write a note on multi-tape Turing machine. CO5 (06)
10. a) Design a Turing machine that accepts $L = \{n_a(W) = n_b(W)\}$. CO5 (10)
 b) Explain the hierarchy of language classes in automata theory with diagram. CO5 (05)
 c) Define Post Correspondence Problem. Solve the following instance of PCP. CO5 (05)

	List A	List B
i	W_i	X_i
1	110	101
2	0011	00
3	0110	110
